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Department of Education
National Capital Region
**SCHOOLS DIVISION OFFICE**
**MARIKINA CITY**

# MATHEMATICS

## Quarter 1: Module 2
## Arithmetic Sequence

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# What I Need to Know

Hello Grade 10 learners! In this module, you will learn how to:

| |
|---|
| **Illustrate an arithmetic sequence. *ML10AL-Ib-1*** |

You can say that you have understood the lesson in this module if you can already;

1. define and illustrate arithmetic sequence
2. determine whether the given sequence is arithmetic or not
3. apply the concept of arithmetic sequence

# What I Know

Read each item carefully**.** Encircle the letter that corresponds to the correct answer.

1. The term of an arithmetic sequence has a common _____.
   A. difference B. factor C. ratio D. term

2. A sequence in which each term is obtained by adding the preceding term by a common difference is known as _____.
   A. arithmetic B. Fibonacci C. geometric D. harmonic

3. What is the next term in the arithmetic sequence 18, 14, 10, 6,…?
   A. 22 B. 4 C. 2 D. 0

4. What is the common difference in item number 3?
   A. - 4 B. -2 C. 2 D. 4

5. What are the next three terms of the arithmetic sequence, -6, -1, 4, 9,…?
   A. 14, 19, 24 B. 16, 18, 20 C. 17, 20, 23 D. 19, 24, 29

6. What are the next two terms in the arithmetic sequence 38, 32, 26, 20,…?
   A. 14, 6 B. 14, 8 C. 14, 10 D. 24, 18

7. Which is the common difference of 2y, 3y + 1, 4y + 2?
   A. y - 1 B. y + 1 C. y + 2 D. 2y +1

8. The eighth, ninth and tenth terms of an arithmetic sequence are 18, 24 and 30 respectively. What is the 18th term?
   A. 54 B. 78 C. 88 D. 102

9. Given the arithmetic sequence $\frac{1}{2}x + 3$, x + 2, $\frac{3}{2}x + 1$, what is the next term?
   A. 2x B. 2x +1 C. $\frac{1}{2}x$-1 D. $\frac{5}{2}x + 2$

10. If the design lengths of the ladders are 14m, 11m, 8m, what is the length of the smallest ladder?
   A. 2m B. 12m C. 14m D. 16m

# ARITHMETIC SEQUENCE

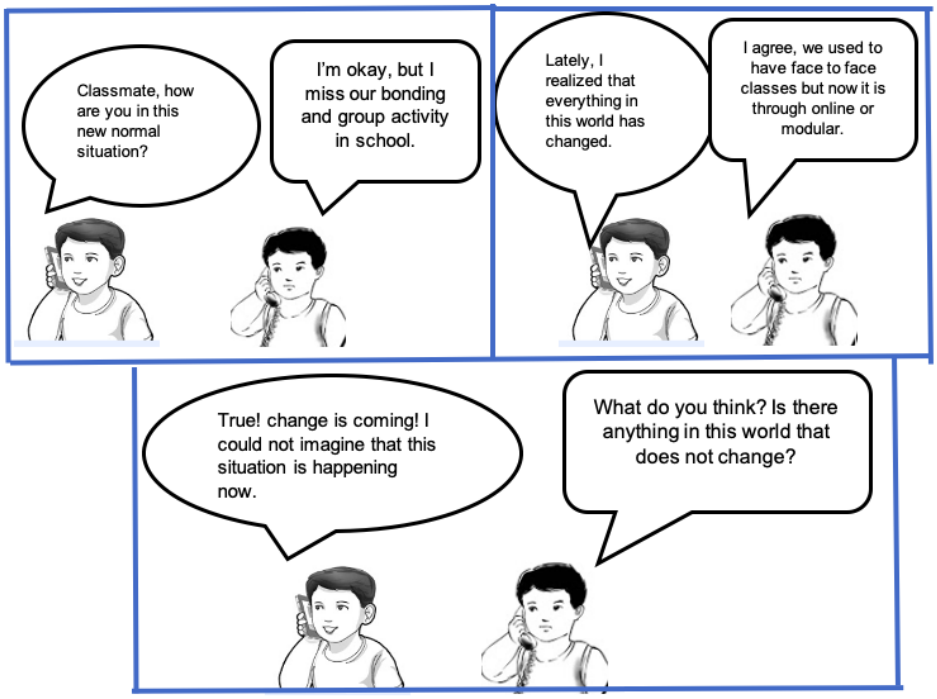
## What's In

A. Find the next 5 terms in the given sequence.
   1. 8,11,14,17,20,…
   2. 4, 10, 16, 22,…
   3. $\frac{13}{3}, \frac{17}{3}, 7, \frac{25}{3}, \ldots$

B. Give 5 examples of things or situations inside your house that illustrate patterns.

## What's New

Read the conversation of the two students.




Answer the question of one student: *Is there anything in this world that does not change?*

If yes, cite an example. _____

If none, cite changes in your life or in your family that you have experienced during community quarantine. _____

## What is it

In this life, change is constant. It means that change is always there. There are changes in our behavior, physical appearance and even in our age. These changes may represent an arithmetic sequence.

An **arithmetic sequence** or **arithmetic progression** is a sequence in which the difference between any two consecutive terms is **constant**, called **common difference**, denoted by ***d***. The concept of arithmetic sequence can be observed in different ways.

Consider the calendar for the month of August 2020 below.

**August 2020**

| Sunday | Monday | Tuesday | Wednesday | Thursday | Friday | Saturday |
|---|---|---|---|---|---|---|
| | | | | | | 1 |
| 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| 16 | 17 | 18 | 19 | 20 | 21 | 22 |
| 23 | 24 | 25 | 26 | 27 | 28 | 29 |
| 30 | 31 | | | | | |

Observe the dates that fall on Monday. These are 3, 10, 17, 24, and 31. The difference between each date is 7 such that :

$$10 - 3 = 7$$
$$17 - 10 = 7$$
$$24 - 17 = 7$$
$$31 - 24 = 7$$

Since the difference is constant, therefore the sequence is an arithmetic sequence with 7 as a common difference.

Can you give another arithmetic sequence from the calendar?

Here are the other of examples of arithmetic sequence from the calendar:

a. 9, 10, 11, 12, 13, 14, 15 *The common difference is 1.*
b. 1, 8, 15, 22, 29 *The common difference is 7.*
c. 2, 10, 18, 26 *The common difference is 8.*

To find the common difference, you can use the formula:

**d =** $\boldsymbol{a_4 - a_3 = a_3 - a_2 = a_2 - a_1}$ or
**d =** $\boldsymbol{a_n - a_{n-1}}$

For example, in the sequence 2, 10, 18, 26, to find the common difference you have:

Let $a_1 = 2, a_2 = 10, a_3 = 18, a_4 = 26$

Then d = $a_2 - a_1$
= 10 – 2
= 8

d = $a_3 - a_2$
= 18 – 10
= 8

d = $a_4 - a_3$
= 26 - 18
= 8

Since the sequence 2, 10, 18, 26 has a common difference of 8, then it is an arithmetic sequence.

If an arithmetic sequence has n terms and the common difference is d, then the sequence $a_1, a_2, a_3, a_4, \dots a_n$ can be written as $a_1, a_1 + d, a_2 + d, a_3 + d, \dots\ a_{n-1} + d$.

Hence, $a_1, a_1 + d, a_2 + d, a_3 + d, \dots\ a_{n-1} + d$.

The arithmetic sequence 2, 10, 18, 26 can also be written as 2, 2 + 8, 10 + 8, 18 + 8.

***Example 1**:* If the 13th , 14th, and 15th terms of an arithmetic sequence are 18, 21, 24 respectively, what is the 19th term?

Solution:

To answer the problem, let's do the steps below.

Step 1: Find the common difference.

d = $a_{14} - a_{13}$
= 21 – 18
= **3**

d = $a_{15} - a_{14}$
= 24 – 21
= **3**

*The common difference is 3.*

Step 2: To find the 19th term, add the common difference to the preceding term to get the next terms.

$a_{13}$ = 18
$a_{14}$ = 18 + **3** = 21
$a_{15}$ = 21 + **3** = 24
$a_{16}$ = 24 + **3** = 27
$a_{17}$ = 27 + **3** = 30
$a_{18}$ = 30 + **3** = 33
$a_{19}$ = 33 + **3** = **36**

*Therefore, the 19th term of the sequence is 36.*

**Example 2:** In arithmetic sequence 21, 19, 17, 15, 13, … , find the common difference.

Solution:

$$d = a_2 - a_1$$
$$= 19 - 21$$
$$\mathbf{d = -2}$$

*Therefore, the common difference of the arithmetic sequence is - 2.*

**Example 3**: Given the sequence $\frac{1}{3}x$+2, $\frac{7}{3}x$+3, $\frac{13}{3}x$+4, find the common difference and the next term of the sequence.

Solution:

To answer the problem, let's do the steps below.

Step 1: Find the common difference.

$$d = a_3 - a_2$$

d = $\frac{13}{3}x$ + 4 - ($\frac{7}{3}x$ + 3) *substitute the value*

d = $\frac{13}{3}x$ + 4 - $\frac{7}{3}x$ – 3 *apply the DPMA/S*

d = $\frac{13}{3}x - \frac{7}{3}x$ + 4 -3 *combine similar terms*

**d = 2x +1**

Step 2: To find the next term, add the common difference to the preceding term.

$$a_4 = a_3 + d$$

$a_4$ = $\frac{13}{3}x$ + 4 + **2x +1** *substitute the value*

= $\frac{13}{3}x + 2x$ + 4 +1 *combine similar terms*

= $\frac{13+6}{3}x$ +5 *LCD for the given fraction*

$$\boldsymbol{a_4 = \frac{19}{3}x + 5}$$

*Therefore, the next term of the sequence is* $\boldsymbol{\frac{19}{3}x + 5}$.

## What's More

**A. Determine whether the given sequence is arithmetic or not. If it is an arithmetic find the common difference.**

1. -6, -4, -1, 3
2. 6, 4, 2, 0, -2
3. 0.3, 1.3, 2.3, 3.3
4. 15, 10, 5, 0, -5, -10
5. $\frac{1}{9}$, $\frac{1}{3}$, 1, 3

**B. Write the first 5 terms of an arithmetic sequence with the given conditions.**

1. $a_1$ = 13, d = -3 _____
2. $a_2$ = 3, d = 1 _____
3. $a_3$ = 9, d = 2 _____
4. $a_1$ = 9, d = $-\frac{3}{4}$ _____
5. $a_2$ = -12, d = 12 _____

## What I Have Learned

Fill in the blanks to complete the statements.

1. A/An _____ is a sequence where every term after the first is obtained by adding a constant.
2. Consecutive terms of an arithmetic sequence have common _____.
3. To find the common difference, you may use the formula _____.
4. In a sequence 5,12,19, 26, … the common difference is _____ and the next term is _____.

## What I Can Do

Give a situation or scenario where you can apply the concept of arithmetic sequence in your daily life. Present the situation/scenario in any of the following: drawing or illustration, tabular, diagram and others. List the first ten terms of your arithmetic sequence and determine the common difference.

RUBRIC FOR THE ACTIVITY:

| Score | Description |
|---|---|
| 10 points | The situation/scenario is completely presented using drawing or illustration, tabular, diagram and others. All the terms and the common difference are correct. |
| 7 points | The situation/scenario is completely presented using drawing or illustration, tabular, diagram and others. At least 7 terms and the common difference are correct. |
| 5 points | The situation/scenario is partially presented using drawing or illustration, tabular, diagram and others. At least 5 terms and the common difference are correct. |
| 3 points | The situation/scenario is not presented using any of the required method. At least 3 terms and the common difference are correct. |
| No points earned | No attempt. |

## Assessment

Encircle the letters which give the correct answers.

1. A sequence in which each term is obtained by adding the preceding term by a common difference is _____.
   A. arithmetic B. Fibonacci C. geometric D. harmonic

2. The term of an arithmetic sequence has a common _____.
   A. term B. ratio C. factor D. difference

3. What is the common difference in the sequence 0, 4, 8, 12,…?
   A. 4 B. 2 C. - 2 D. - 4

4. Given the arithmetic sequence -4, -1, 2, 5,.. what is the next term?
   A. 10 B. 8 C. 6 D. 4

5. Find the next three terms of the given arithmetic sequence -6, -1, 4, 9,…
   A. 14, 19, 24 B. 16, 18, 20 C. 17, 20, 23 D. 19, 24, 29

6. Find the 10th term in the sequence 38, 32, 26, 20, …
   A. -104 B. -16 C. 16 D. 104

7. Find the common difference of 2y, 3y +1, 4y + 2.
   A. 2y + 1 B. y + 2 C. y + 1 D. y - 1

8. The 8th, 9th and 10th terms of an arithmetic sequence are 18, 24 and 30, respectively. What is the 22nd term?
   A. 72 B. 84 C. 96 D. 102

9. In the arithmetic sequence $\frac{1}{2}x + 3$, x + 2, $\frac{3}{2}x + 1$, what is the next term?
   A. 2x B. 2x +1 C. $\frac{1}{2}x + 2$ D. $\frac{5}{2}x$ - 1

10. What is the common difference in number 9?
   A. 2x B. 2x + 1 C. $\frac{1}{2}x$ - 1 D. $\frac{5}{2}x + 2$

## *Additional Activities*

**A. Answer the problem below.**

Donalene is an employee of a good company in the Philippines. Last December, she received an amount of ₱100,000.00 from the company as her bonus and incentive pay. Donalene decided to save ₱90,000.00 for her monthly house rental amounting to P10,000.00 every end of the month.

1. If Donalene will pay the monthly rental using her savings, what month will her account be in zero balance?
2. What are the first three terms in the sequence?
3. How much money will be left from her savings at the end of June?

**B. Answer the given problems below. Choose the letter that corresponds to the answer of each item and write it in the box to decode the word.**

**Given**

1. In the sequence 0 ,-5, -10, -15, what is the 3rd term?
2. If $a_2 = -3$ , d= 4, what is the 1st term?
3. If $a_1 = 4$ , d = -3, what is the 5th term?
4. If $a_2 = 18,\ d = -2$, write the first 5 terms.
5. Given the first four terms 7,14, 21, 28,… , what is the 6th term?
6. Given the sequence 6, 10, 14, 18,… , what is the next term?
7. Given the first four terms -14, -7, 0, 7, 14,… , what is *d*?
8. If d=7, the first term is 52, what is the 3rd term?
9. Given the sequence 2 + x, 3 + 3x, 4 + 5x,.. what is the next term?
10. Given the sequence x, 6x, 11x, 16x, what is the common difference?
11. Given the sequence 3x + 2, -x + 5, -5x +8,… , find the next 2 terms.
12. List down the first 5 terms of an arithmetic sequence, whose 5th term is 27 and d = -3.

**Choices**

| | |
|---|---|
| A | 20, 18, 16, 14, 12 |
| C | -8 |
| D | 42 |
| E | 39, 36, 33, 30, 27 |
| F | -10 |
| G | -9x +11, -13x +14 |
| H | 66 |
| I | 7 |
| L | 22 |
| R | -7 |
| S | 5x |
| U | 5 + 7x |

Question: He is the Prince of Mathematics. Who is he?

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | |
| 3 | 4 | 2 | 6 | 1 | 2 | 7 | 12 | 5 | 2 | 7 | 3 | 8 | 11 | 4 | 9 | 10 | 10 |

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